



TSF in a nutshell

Functional safety in the context of an Open Source based ecosystem

Kaspar Mätas, *Codethink*

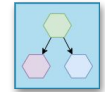
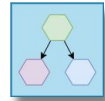
Daniel Krippner, *ETAS GmbH*

What is the Trustable Software Framework (TSF)?



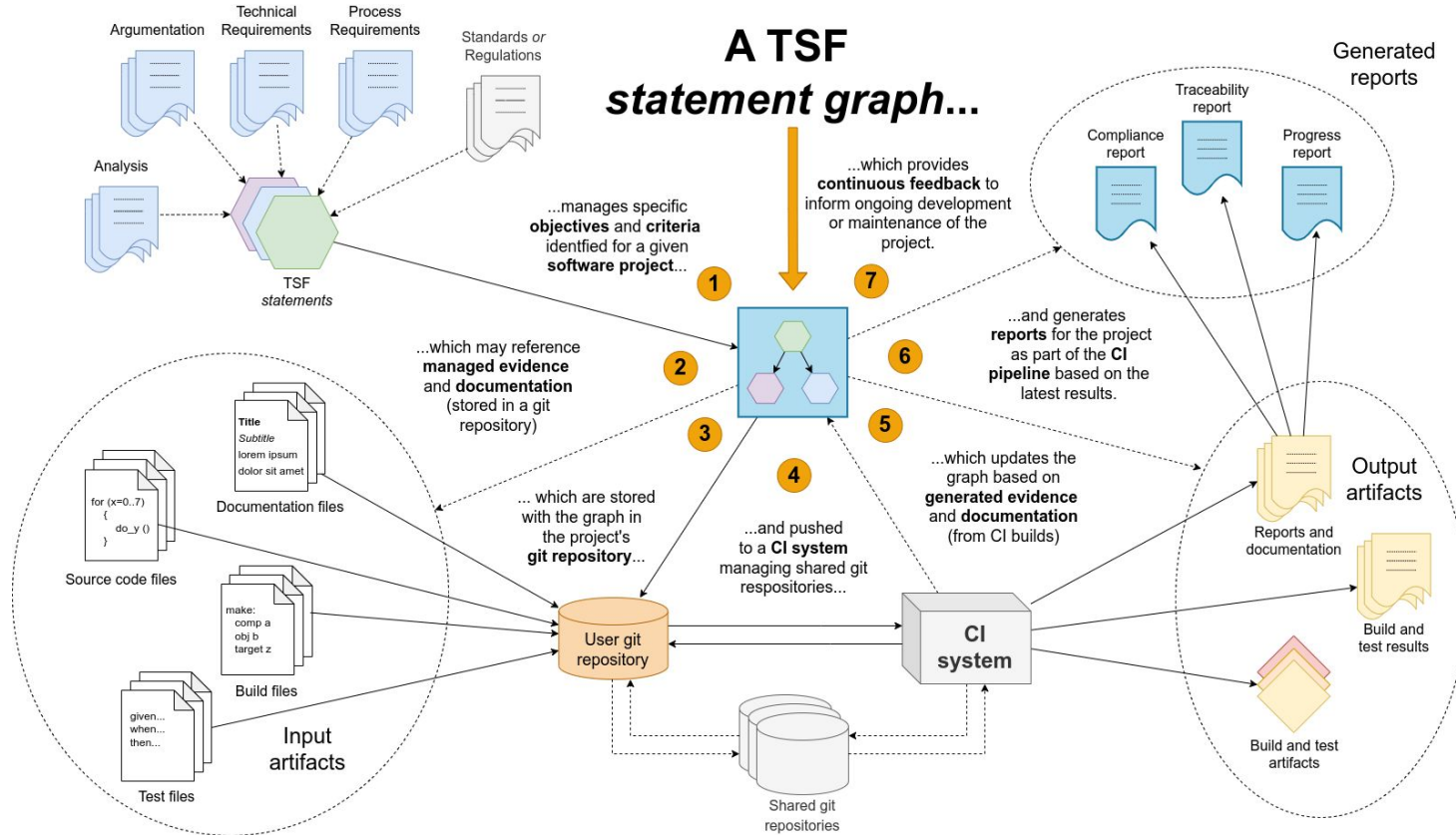
As a framework, TSF needs to provide the following:

- **Objectives:** defining what is important, and what it is trying to accomplish.¹
- **Model:** a simplified description of a more complex system or idea, focusing on specific elements or relationships.
- **Methodology:** a system of methods used for a particular activity, which may use models
- **Tools:** to help apply these methods, which are adaptable to users' needs and project-specific objectives.
- **Documentation:** to provide guidance and examples of how to use the framework.



¹ See the *Towards Trustable Software* white paper available from <https://www.trustablesoftware.com/>

TSF: iterative and construction-driven



exida Functional Safety Assessment for TSF

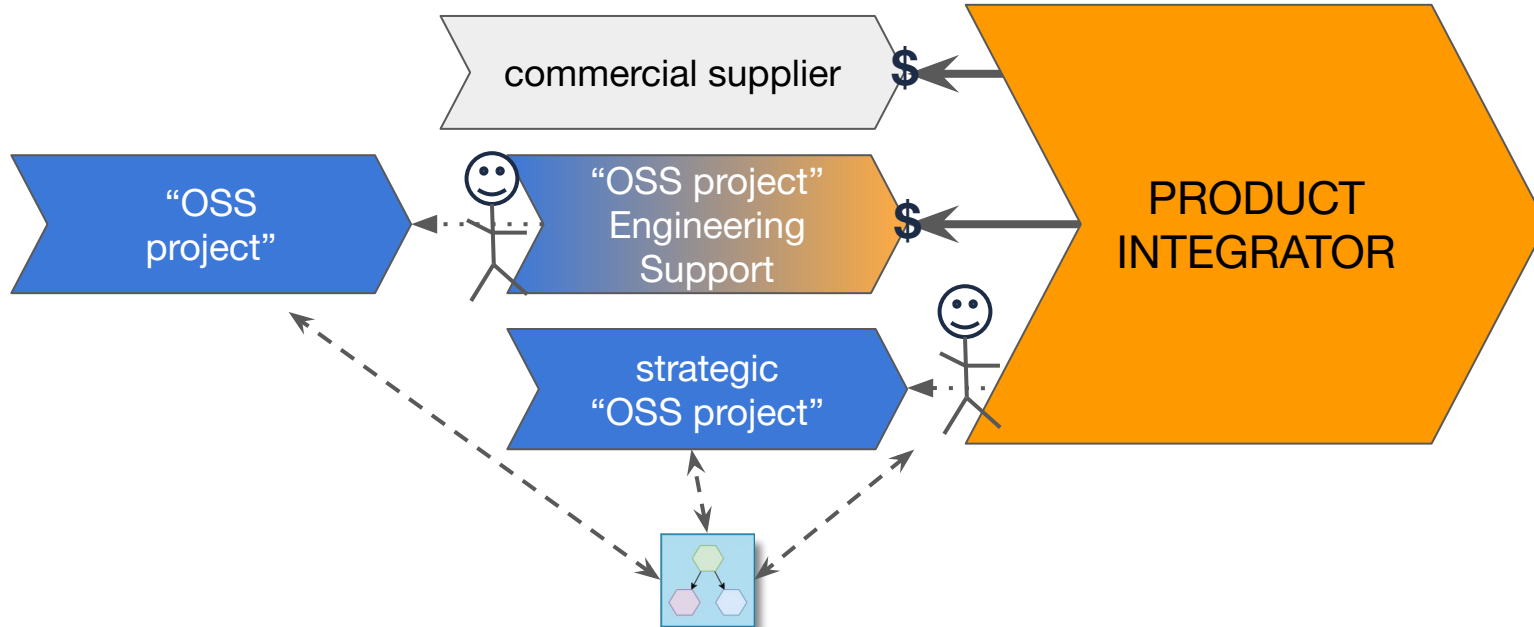


"The functional safety assessment was performed to the requirements of the applicable parts of IEC 61508 with experts knowledgeable in ISO 26262 ensuring compatibility of the activities with the specific additional requirements and requests of the automotive industry. The results of the functional safety assessment can be summarized as follows:

Considering the process objectives of both IEC 61508 and ISO 26262 the specification of Trustable and the reference implementation that maps trustable tenets and assertions to the objectives in IEC 61508 **it is determined that Trustable sets out an approach to software development which meets and in many cases exceeds the rigor and expectations established for software in IEC 61508 at SIL 3.**

Organizations that attempt to implement TSF according to the reference model may proceed confidently that they will be able to measure their progress towards a future certification attempt safe in the knowledge that the requirements for a successful assessment will be met."

Why is TSF great for OSS development models?



Openly extend&evolve TSF extension libraries

TSF Engineering&Artifacts can flow upstream



**How to apply TSF in the open?
Come and get in touch**